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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I Maithreyan.P (192722600089) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: .

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10ma
Understanding of Scenario	20%	2	2	2	2
Identification of Key Issues	20%	2	2	1	1
Application of Concepts	25%	1	2	1	1
Decision Making	20%	1	1	1	1
Clarity of Response	15%	1	1	1	1

28/30

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Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

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2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

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		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand

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Subnetting, IPv6 and Routing

Scenario-Based Assessment: Answers

Scenario 1: Smart Campus (192.168.100.0/24)

Approach: Allocate the largest subnet first so the blocks stay aligned: /25 (128 addresses), /26 (64), then /27 (32). Usable hosts = 2^(32 − prefix) − 2.

Department	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
School of Computing	192.168.100.0/25	192.168.100.0	192.168.100.127	192.168.100.1 – 192.168.100.126	126
School of Electronics	192.168.100.128/26	192.168.100.128	192.168.100.191	192.168.100.129 – 192.168.100.190	62
School of Mechanical Engineering	192.168.100.192/27	192.168.100.192	192.168.100.223	192.168.100.193 – 192.168.100.222	30

Unused range: 192.168.100.224 – 192.168.100.255 (192.168.100.224/27, 32 addresses), available for future growth.

Scenario 2: Multinational Software Company (10.10.0.0/16)

Approach: Allocate in order of size from the start of the block: /24, /25, /26, /27.

Department	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	10.10.0.0/24	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	10.10.1.0/25	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	10.10.1.128/26	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	10.10.1.192/27	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

Unused range: 10.10.1.224 through 10.10.255.255. This is the rest of 10.10.1.0/24 (10.10.1.224/27) plus all of 10.10.2.0 – 10.10.255.255.

Check: Total block = 65,536 addresses. Used = 256 + 128 + 64 + 32 = 480. Unused = 65,056 addresses.

Scenario 3: Smart Hospital IPv6 Migration

Given: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. **Compressed notation:** Drop leading zeros in each group (0db8 becomes db8), then replace the longest run of all-zero groups with "::". Result: 2001:db8:abcd::/64

2. **Expanded (full) notation:** 2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. **Network prefix (/64):** The first 64 bits (the first four groups) identify the network: 2001:0db8:abcd:0000::/64, written as 2001:db8:abcd:0::/64. The remaining 64 bits are the interface identifier (host portion).

4. **Host addresses supported in the subnet:**
$2^{(128 - 64)} = 2^{64} =$ 18,446,744,073,709,551,616 addresses.

5. **Total possible host addresses in this network:** Also 2^64 = 18,446,744,073,709,551,616. IPv6 has no broadcast address, so every address can be assigned to a host. Only the all-zeros address (subnet-router anycast) is conventionally reserved, leaving 2^64 − 1 assignable in practice.

Scenario 4: ISP Core Router Forwarding

Destination IP: 192.168.2.45

Routing Table Entry	Address Range	Match?
192.168.1.0/24	192.168.1.0 – 192.168.1.255	No
192.168.2.0/24	192.168.2.0 – 192.168.2.255	Yes
192.168.3.0/24	192.168.3.0 – 192.168.3.255	No

1. **Destination subnet:** 192.168.2.0/24 (network 192.168.2.0, broadcast 192.168.2.255).

2. **Matching routing table entry:** 192.168.2.0/24. The first three octets match (192.168.2) and it is the longest matching prefix.

3. **Next hop / interface:** The packet is forwarded out the router interface connected to the 192.168.2.0/24 customer network (or to the next-hop address configured for that entry).

4. **Default route:** 0.0.0.0/0, pointing to the ISP's upstream router. Any packet matching none of the three entries is sent there. If no default route exists, the packet is dropped.

Scenario 5: Two LANs Connected by a Layer-3 Router

Item	Administration LAN	Finance LAN
Subnet	192.168.10.0/26	192.168.10.64/26
Subnet mask	255.255.255.192	255.255.255.192
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62
Gateway (router interface)	192.168.10.1	192.168.10.65

Suggested computer addresses (the gateway address is excluded):

Administration LAN	Finance LAN

PC1: 192.168.10.10	PC1: 192.168.10.74

Administration LAN	Finance LAN
PC2: 192.168.10.11	PC2: 192.168.10.75
PC3: 192.168.10.12	PC3: 192.168.10.76

Default gateways: Hosts in the Administration LAN use 192.168.10.1. Hosts in the Finance LAN use 192.168.10.65. Each gateway is the router interface in the host's own subnet, which is how traffic between the two LANs is routed.